Table 1: Factors affecting public and private blockchain platforms

Factors	Purpose	Public blockchain platform	Private blockchain platform
Confidentiality	Limits the entities participating in a transaction	Low	High
Throughput	Number of transactions processed end-to-end for a given time period	Low	High
Anonymity of users	Closed user group for accessing the transaction	Low	High
Finality	Time to wait to guarantee transaction is not orphaned (irreversible)	Low (Lottery based algorithm to select a node for transaction)	High (Voting based consensus algorithm to select node for transaction)
Access to platform	Getting entry to the platform either anonymously or based on invitation from members	High (Anyone can access)	Low (Invitation based access)

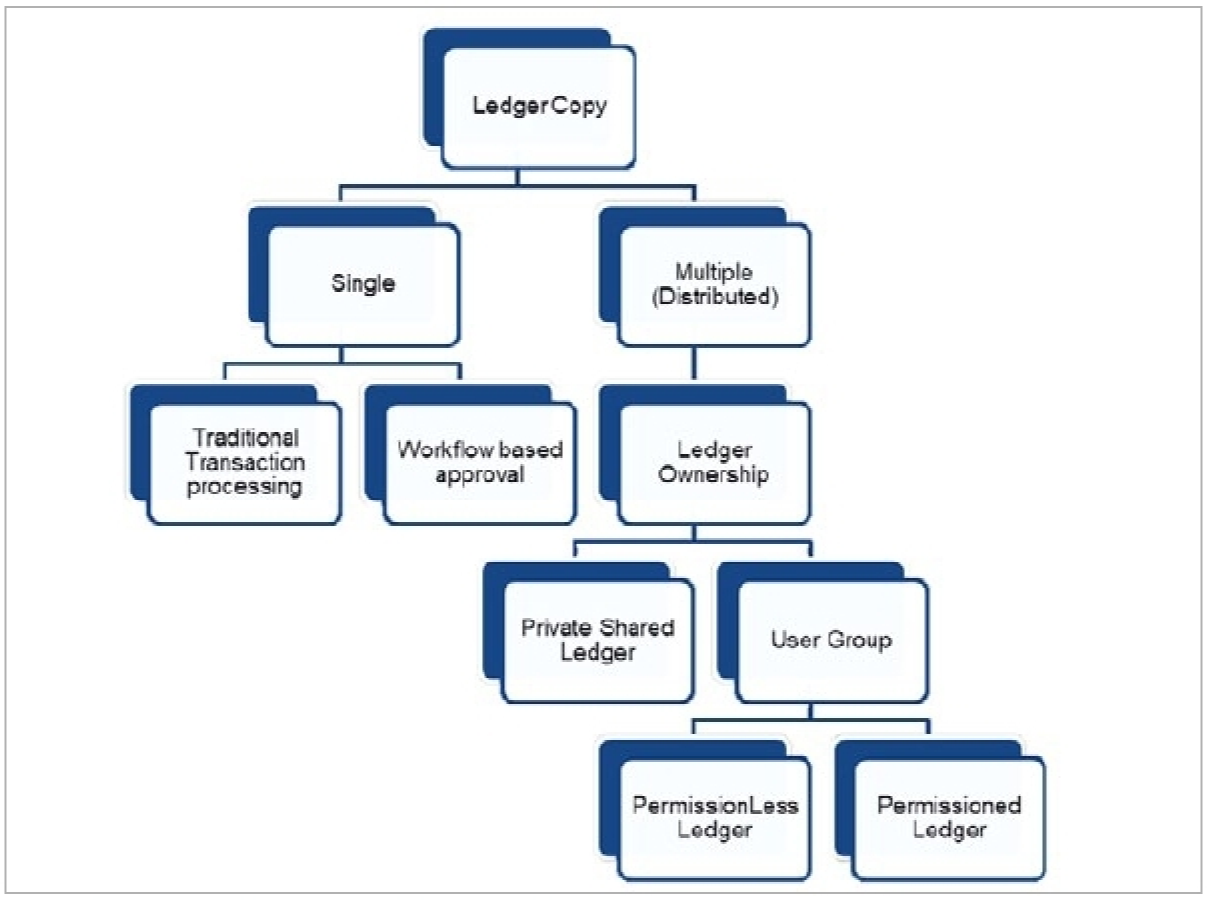

Figure 3: Permissioned vs permissionless ledger hierarchy

for a transaction. A public blockchain uses a lottery based algorithm and hence is slower in deciding a block. A private blockchain uses a voting based algorithm where the majority votes decide whether the block is to be finalised and hence is faster in deciding a block for transaction.

Credibility: Public blockchain platforms are trustless. The participants in the platform network may not trust each other as they work anonymously. On the other side, the private blockchain is a closed set of users and hence the credibility of the authorised nodes is high.

Collision of transaction: In a private blockchain, since the work group is known to each other, it is less likely to have a collision of transactions as the block of transaction handling is clearly defined for each user in the network. On the other hand, a public blockchain is an open-ended network and there are many collisions of transactions possible as multiple users try to attack a transaction, as no one in the network knows who validates the transaction.

Though the blockchain is widely used for money related transactions, there are many proven use cases across different industries like retail (for stock checking and order management), in manufacturing (for goods movement and processing), and in healthcare (for patient database management) to name a few. With this in mind, it is always important to understand the merits and demerits of both public and private blockchains prior to deciding which one is more suitable for an industry-specific use case.

Since the decentralised nature of the blockchain and its security are of key concern in implementing a blockchain based platform, it is important for an enterprise architect to choose one of the open or closed blockchain platform solutions or even a hybrid platform where some implementation (or workflow) is open and some implementation (or workflow) is closed. A hybrid blockchain platform sometimes makes the platform more flexible to adapt to either the private blockchain or public blockchain paradigm. END

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